

Problem 1

A fair coin is tossed repeatedly. Show that the following two statements are equivalent:

- (a) the outcomes of different tosses are independent,
- (b) for any given finite sequence of heads and tails, the chance of this sequence occurring in the first m tosses is 2^{-m} , where m is the length of the sequence.

Proof.

□

Problem 2

A symmetric random walk takes place on the integers $0, 1, 2, \dots, N$ with absorbing barriers at 0 and N , starting at k . Show that the probability that the walk is never absorbed is zero.

Proof.

□

Problem 3

Suppose $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $B \in \mathcal{F}$ satisfies $\mathbb{P}(B) > 0$. Let $\mathbb{Q} : \mathcal{F} \rightarrow [0, 1]$ be defined by $\mathbb{Q}(A) = \mathbb{P}(A \mid B)$. Show that $(\Omega, \mathcal{F}, \mathbb{Q})$ is a probability space. If $C \in \mathcal{F}$ and $\mathbb{Q}(C) > 0$, show that $\mathbb{Q}(A \mid C) = \mathbb{P}(A \mid B \cap C)$; discuss.

Proof.

□

Problem 4

A biased coin is tossed repeatedly. Each time there is a probability p of a head turning up. Let p_n be the probability that an even number of heads has occurred after n tosses (zero is an even number). Show that $p_0 = 1$ and that $p_n = p(1 - p_{n-1}) + (1 - p)p_{n-1}$ if $n \geq 1$. Solve this difference equation.

Proof.

□

Problem 5

Loaded dice.

- (a) Show that it is not possible to weight two dice in such a way that the sum of the two numbers shown by these loaded dice is equally likely to take any value between 2 and 12 (inclusive).

(b) Given a fair die and a loaded die, show that the sum of their scores, modulo 6, has the same distribution as a fair die, irrespective of the loading.

Proof.

□